MIL-T-36093A(BuMed)
3 April 1963
SUPERSEDING
MIL-T-36093(BuMed)
26 August 1961

MILITARY SPECIFICATION

TEST SET, BINOCULAR VISION (WORTH)

1. SCOPE

1.1 This specification covers a binocular vision test set to be used for a rapid determination of the absence or presence of binocular vision.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

W-F-421	Flashlights, Electric, Hand.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-665	Boxes; Paperboard, Metal Stayed (Including Stay Material).

Military

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSN 6515

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DRAWING

Defense Medical Supply Center

20449 Test Set, Binocular Vision, Worth.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Material

3.1.1 Flashlight. The flashlight shall be fabricated of the materials specified in Specification W-F-421.

3.1.2 Filter assembly. The body of the filter assembly shall be fabricated from a suitable rigid black (opaque) plastic. The filters shall be fabricated of plastic. The red and green filters of the filter assembly and the red and green filters of the spectacles shall be complementary colors (see 3.3). The diffusion plate shall be fabricated of a colorless translucent plastic.

3.1.3 Spectacles. The frames shall be fabricated from nickel silver or other suitable corrosion resistant metal. The filters shall be of plastic and as indicated in 3.1.2.

3.2 Style, design and dimensions. The test set shall be of the Worth four dot test type for the rapid determination of the absence or presence of binocular vision. The set shall consist of a flashlight with filter assembly insert and a pair of red-green spectacles. The unit shall be supplied without batteries.

3.2.1 Flashlight. The flashlight shall be in accordance with the requirements of Specification W-F-421 for class 1 or 2, size 1 (two cell).

3.2.2 Filter assembly. The filter assembly shall be in accordance with Drawing 20449. The filter assembly shall be inserted under the flashlight cap and lens as shown on the drawing.

3.2.3 Spectacles. The spectacles shall be constructed as shown on the Drawing, with one red filter and one green filter inserted in the frames. The X bridge shall permit the wearer to reverse the spectacles.

3.3 Performance. When tested as specified in 4.4.1 the following observations shall be made:

(1) When viewing through both filters of the spectacles, the four dots shall be visible.

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(2) When viewing through the red filter of the spectacles only, two dots shall be visible (one white and one red).

(3) When viewing through the green filter of the spectacles only, three dots shall be visible (one white and two green).

3.4 Finish. The finish on the flashlight shall be as specified in Specification W-F-421. The spectacle frames shall have a smooth bright finish.

3.5 Identification marking. The test set shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. Tests sets shall be free from defects which detract from their appearance or may impair their serviceability.

3.7 Instructions. Instructions for use shall be included with each set.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facility or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with Standard MIL-STD-105.

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4.2.2 Sampling for examination. A random sample of test sets shall be selected from each lot offered to the Government in accordance with Standard MIL-STD-105. Examination for dimensions shall be in accordance with Standard MIL-STD-105, and as indicated in table I.

4.2.3 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105, and as indicated in table I.

Table I. Sampling Criteria

Component	Characteristic	Rqmt.	Test Procedure	Unit of Product	Insp. level	AQL (percent defective)
Test Set	Performance	3.3	4.4.1	1 set	L3*	**
	Dimensions	3.2		1 set	L3	4.0

* A minimum of three units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

4.2.4 Components. Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the test sets to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The test sets shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Performance test. The filter assembly shall be placed in the flashlight as indicated on the Drawing. Two fresh batteries (size D) shall be inserted in the flashlight case. With the examiner wearing the spectacles and the flashlight turned on, the filter assembly shall be viewed from a distance of approximately 13 inches. The first observation shall be made with both eyes open; the second observation shall be made through the red filter of the spectacles only (by closing the eye with the green filter); the third observation shall be made through the green filter of the spectacles only (by closing the eye with the red filter). The test shall be performed by a person having normal binocular vision.

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5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Unit of issue. One (1) test set as specified constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit and intermediate package, and exterior container, as applicable, for the required level of protection specified in procurement document.

Table II. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-636 or PPP-B-665. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636. Commercial colors shall be acceptable when carton conforming to Federal Specification PPP-B-566 or PPP-B-676 is utilized. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III.

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5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for type 2 load and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with Standard MIL-STD-129, contract number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129, except as follows: The letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

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6.3 Transportation description. Transportation description applicable to this item is:

Optical Goods or Instruments, NOIBN
Carload minimum weight 10,000 pounds subject to Rule 34 of the Uniform Freight Classification.
Truckload minimum weight 10,000 pounds per National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army-Med
Navy-BuMed
Air Force-AFCSG

Preparing activity:
Defense Medical Supply Center

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